Exercise 4.3

1. Find HCF by factorization method.

(i) $\boldsymbol{21x^2y, 35xy^2}$

$$21x^2y = 3 \cdot \boxed{7} \cdot \boxed{x} \cdot x \cdot \boxed{y}$$
$$35xy^2 = 5 \cdot \boxed{7} \cdot \boxed{x} \cdot y \cdot \boxed{y}$$
$$\boldsymbol{HCF = 7xy}$$

(ii) $\boldsymbol{4x^2 - 9y^2, 2x^2 - 3xy}$

Factorization

$$\begin{aligned} & 4x^2 - 9y^2 \\ &= (2x)^2 - (3y)^2 \\ &= (2x + 3y)(2x - 3y) \end{aligned}$$

$$\begin{aligned} & 2x^2 - 3xy \\ &= x(2x - 3y) \end{aligned}$$

Now

$$4x^2 - 9y^2 = (2x + 3y)\boxed{(2x - 3y)}$$
$$2x^2 - 3xy = x\boxed{(2x - 3y)}$$
$$\boldsymbol{HCF = (2x - 3y)}$$

(iii) $\boldsymbol{x^3 - 1, x^2 + x + 1}$

Factorization

$$\begin{aligned} & x^3 - 1 \\ &= (x)^3 - (1)^3 \\ &= (x + 1)[(x)^2 + (x)(1) + (1)^2] \\ &= (x + 1)(x^2 + x + 1) \end{aligned}$$

Now

$$x^3 - 1 = (x + 1)\boxed{(x^2 + x + 1)}$$
$$x^2 + x + 1 = \boxed{x^2 + x + 1}$$
$$\boldsymbol{HCF = x^2 + x + 1}$$

(iv) $\boldsymbol{a^3 + 2a^2 - 3a, 2a^3 + 5a^2 - 3a}$

Factorization

$$\begin{aligned} & a^3 + 2a^2 - 3a \\ &= a[a^2 + 2a - 3] \\ &= a[a^2 + 3a - a - 3] \\ &= a[a(a + 3) - 1(a + 3)] \\ &= a[(a + 3)(a - 1)] \\ &= a(a + 3)(a - 1) \end{aligned}$$

Also

$$\begin{aligned} & 2a^3 + 5a^2 - 3a \\ &= a[2a^2 + 5a - 3] \\ &= a[2a^2 + 6a - a - 3] \\ &= a[2a(a + 3) - 1(a + 3)] \\ &= a[(a + 3)(2a - 1)] \\ &= a(a + 3)(2a - 1) \end{aligned}$$

Now

$$a^3 + 2a^2 - 3a = \boxed{a(a + 3)}(a - 1)$$
$$2a^3 + 5a^2 - 3a = \boxed{a(a + 3)}(2a - 1)$$
$$\boldsymbol{HCF = a(a + 3)}$$

(v) $\boldsymbol{t^2 - 3t - 4, t^2 + 5t + 4, t^2 - 1}$

Factorization

$$\begin{aligned} & t^2 - 3t - 4 \\ &= t^2 - 4t + t - 4 \\ &= t(t - 4) + 1(t - 4) \\ &= (t - 4)(t + 1) \end{aligned}$$

Also

$$\begin{aligned} & t^2 + 5t + 4 \\ &= t^2 + 4t + t + 4 \\ &= t(t + 4) + 1(t + 4) \\ &= (t + 4)(t + 1) \end{aligned}$$

Also

$$\begin{aligned} & t^2 - 1 \\ &= (t)^2 - (1)^2 \\ &= (t + 1)(t - 1) \end{aligned}$$

Now

$$t^2 - 3t - 4 = (t - 4)\boxed{(t + 1)}$$
$$t^2 + 5t + 4 = (t + 4)\boxed{(t + 1)}$$
$$t^2 - 1 = \boxed{(t + 1)}(t - 1)$$
$$\boldsymbol{HCF = t + 1}$$

(v) $\boldsymbol{x^2 + 15x + 56, x^2 + 5x - 24, x^2 + 8x}$

Factorization

$$\begin{aligned} & x^2 + 15x + 56 \\ &= x^2 + 8x + 7x + 56 \\ &= x(x + 8) + 7(x + 8) \\ &= (x + 8)(x + 7) \end{aligned}$$

Also

$$\begin{aligned} & x^2 + 5x - 24 \\ &= x^2 + 8x - 3x - 24 \\ &= x(x + 8) - 3(x + 8) \\ &= (x + 8)(x - 3) \end{aligned}$$

Also

$$\begin{aligned} & x^2 + 8x \\ &= x(x + 8) \end{aligned}$$

Now

$$x^2 + 15x + 56 = \boxed{(x + 8)}(x + 7)$$
$$x^2 + 5x - 24 = \boxed{(x + 8)}(x - 3)$$
$$x^2 + 8x = x\boxed{(x + 8)}$$
$$\boldsymbol{HCF = x + 8}$$

2. Find HCF he following expressions by using division method:

(i) $27x^3+9x^2-3x-10, 3x-2$

$$\begin{array}{r|l} & 9x^2+9x+5 \\ \hline 3x-2 & 27x^3+9x^2-3x-10 \\ & \pm 27x^3 \mp 18x^2 \\ \hline & 27x^2-3x-10 \\ & \pm 27x^2 \mp 18x \\ \hline & 15x-10 \\ & \pm 15x \mp 10 \\ \hline & 0 \end{array}$$

$$\boldsymbol{HCF = 3x-2}$$

(ii) $x^3-9x^2+23x-15, x^2-4x+3$

$$\begin{array}{r|l} & x-5 \\ \hline x^2-4x+3 & x^3-9x^2+21x-9 \\ & \pm x^3 \mp 4x^2 \pm 3x \\ \hline & -5x^2+18x-9 \\ & \mp 5x^2 \pm 20x \mp 15 \\ \hline & -2x+6 \end{array}$$

$$-2x+6=-2(x-3)$$

We ignore -2 because it is not common to both the given polynomials

$$\begin{array}{r|l} & x-1 \\ \hline x-3 & x^2-4x+3 \\ & \pm x^2 \mp 3x \\ \hline & -x+3 \\ & \mp x \pm 3 \\ \hline & 0 \end{array}$$

$$\boldsymbol{HCF = x-3}$$

(iii) $2x^3+2x^2+2x+2, 6x^3+12x^2+6x+12$

$$2x^3+2x^2+2x+2=2(x^3+x^2+x+1)$$
$$6x^3+12x^2+6x+12=6(x^3+2x^2+x+2)$$
$$=2\times 3(x^3+2x^2+x+2)$$

$$\begin{array}{r|l} & 1 \\ \hline x^3+x^2+x+1 & x^3+2x^2+x+2 \\ & \pm x^3 \pm x^2 \pm x \pm 1 \\ \hline & x^2 \quad +1 \end{array}$$

Now

$$\begin{array}{r|l} & x+1 \\ \hline x^2+1 & x^3+x^2+x+1 \\ & \pm x^3 \quad \pm x \\ \hline & x^2 \quad +1 \\ & \pm x^2 \quad \pm 1 \\ \hline & 0 \end{array}$$

$$\boldsymbol{HCF = 2(x^2+1)}$$

(iv) $2x^3-4x^2-16x, x^3-4x, 3x^2-6x$

$$\begin{array}{r|l} & 2 \\ \hline x^3-4x & 2x^3-4x^2-16x \\ & \pm 2x^3 \mp 8x^2 \\ \hline & 4x^2-16x \end{array}$$

$$4x^2-16=4(x^2-4)$$

We ignore 4 because it is not common to both the given polynomials

$$\begin{array}{r|l} & x \\ \hline x^2-4 & x^3-4x \\ & \pm x^3 \mp 4x \\ \hline & 0 \end{array}$$

Now

$$\begin{array}{r|l} & 3 \\ \hline x^2-4 & 3x^2-6x \\ & \pm 3x^2 \quad \mp 12 \\ \hline & -6x+12 \end{array}$$

$$-6x+12=-6(x-2)$$

We ignore -6 because it is not common to both the given polynomials

$$\begin{array}{r|l} & x+2 \\ \hline x-2 & x^2-4 \\ & x^2 \quad \mp 2x \\ \hline & 2x-4 \\ & \pm 2x \mp 4 \\ \hline & 0 \end{array}$$

$$\boldsymbol{HCF = x-2}$$

3. Find LCM of the following expressions by using prime factorization method:

(i) $2a^2b, 4ab^2, 6ab$

$$2a^2b=\boxed{2}\cdot a\cdot\boxed{a}\cdot\boxed{b}$$
$$4ab^2=\boxed{2}\cdot 2\cdot\boxed{a}\cdot\boxed{b}\cdot b$$
$$6ab=\boxed{2}\cdot 3\cdot\boxed{a}\cdot\boxed{b}$$
$$LCM=[2\cdot a\cdot b][2\cdot 3\cdot a\cdot b]$$
$$\boldsymbol{LCM = 12a^2b^2}$$

(ii) x^2+x, x^3+x^2

Factorization

$$x^2+x$$
$$=x(x+1)$$

$$x^3+x^2$$
$$=x^2(x+1)$$

Now

$$x^2+x=\boxed{x}\cdot\boxed{(x+1)}$$
$$x^3+x^2=\boxed{x}\cdot x\cdot\boxed{(x+1)}$$
$$LCM=[x(x+1)][x]$$
$$\boldsymbol{LCM=z^2(x+1)}$$

(iii) a^2-4a+4, a^2-2a

Factorization

$$a^2-4a+4$$
$$=a^2-2a-2a+4$$
$$=a(a-2)-2(a-2)$$
$$=(a-2)(a-2)$$

$$a^2-2a$$
$$=a(a-2)$$

Now

$$a^2-4a+4=\boxed{(a-2)}(a-2)$$
$$a^2-2a=a\boxed{(a-2)}$$
$$LCM=[(a-2)][a\cdot(a-2)]$$
$$\boldsymbol{LCM=a(a-2)^2}$$

(iv) x^4-16, x^3-4x

Factorization

$$x^4-16$$
$$=(x^2)^2-(4)^2$$
$$=(x^2+4)(x^2-4)$$
$$=(x^2+4)[(x)^2-(2)^2]$$
$$=(x^2+4)(x+2)(x-2)$$

$$x^3-4x$$
$$=x(x^2-4)$$
$$=x[(x)^2-(2)^2]$$
$$=x(x+2)(x-2)$$

Now

$$x^4-16=(x^2+4)\boxed{(x+2)(x-2)}$$
$$x^3-4x=x\boxed{(x+2)(x-2)}$$
$$LCM=[(x+2)(x-2)][x\cdot(x^2+4)]$$
$$LCM=x(x+2)(x-2)(x^2+4)$$
$$LCM=x[(x)^2-(2)^2](x^2+4)$$
$$LCM=x(x^2-4)(x^2+4)$$
$$LCM=x[(x^2)^2-(4)^2]$$
$$\boldsymbol{LCM=x(x^4-16)}$$

(v) $16-4x^2, x^2+x-6, 4-x^2$

Factorization

$$16-4x^2$$
$$=4[4-x^2]$$
$$=4[(2)^2-(x)^2]$$
$$=4[(2+x)(2-x)]$$
$$=4(2+x)(2-x)$$

$$x^2+x-6$$
$$=x^2+3x-2x-6$$
$$=x(x+3)-2(x+3)$$
$$=(x+3)(x-2)$$
$$=-(x+3)(2-x)$$

$$4-x^2$$
$$=(2)^2-(x)^2$$
$$=(2+x)(2-x)$$

Now

$$16-4x^2=4\boxed{(2+x)}\boxed{(2-x)}$$
$$x^2+x-6=-(x+3)\boxed{(2-x)}$$
$$4-x^2=\boxed{(2+x)}\boxed{(2-x)}$$
$$LCM=[(2-x)(2+x)][-4\cdot(x+3)]$$
$$LCM=-4(2-x)(2+x)(x+3)$$
$$LCM=-4[(2)^2-(x)^2](x+3)$$
$$LCM=-4(4-x^2)(x+3)$$
$$\boldsymbol{LCM=4(x^2-4)(x+3)}$$

4. The HCF of two polynomials is $y-7$ and their LCM is $y^3-10y^2+11y+70$. If one of the polynomials is $y^2-5y-14$, find the other.

$$HCF=y-7$$
$$LCM=y^3-10y^2+11y+70$$
$$p(y)=y^2-5y-14$$

Let $q(y)$ be the other polynomial

By using formula

$$p(y)\times q(y)=HCF\times LCM$$
$$q(y)=\frac{HCF\times LCM}{p(y)}$$
$$q(y)=\frac{(y-7)(y^3-10y^2+11y+70)}{y^2-5y-14}$$

$$\begin{array}{r|l} & y-5 \\ y^2-5y-14 & y^3-10y^2+11y+70 \\ & \pm y^3\mp 5y^2\mp 14y \\ \hline & -5y^2+15y+70 \\ & \mp 5y^2\pm 15y\pm 70 \\ \hline & 0 \end{array}$$

$$q(y)=(y-7)(y-5)$$
$$q(y)=y(y-5)-7(y-5)$$
$$q(y)=y^2-5y-7y+35$$

$$q(y) = y^2 - 12y + 35$$

5. LCM and HCF of two polynomials $p(x)$ and $q(x)$ are $36x^3(x+a)(x^3-a^3)$ and $x^2(x-a)$ respectively. If $p(x) = 4x^2(x^2-a^2)$, find $q(x)$.

$$LCM = 36x^3(x+a)(x^3-a^3)$$
$$HCF = x^2(x-a)$$
$$p(x) = 4x^2(x^2-a^2)$$
$$q(x) = ?$$

By using formula

$$p(x) \times q(x) = LCM \times HCF$$
$$q(x) = \frac{LCM \times HCF}{p(x)}$$
$$q(x) = \frac{[36x^3(x+a)(x^3-a^3)][x^2(x-a)]}{4x^2(x^2-a^2)}$$
$$q(x) = \frac{36x^3 \cdot x^2(x+a)(x^3-a^3)(x-a)}{4x^2(x+a)(x-a)}$$
$$\boldsymbol{q(x) = 9x^3(x^3-a^3)}$$

6. The HCF and LCM of two polynomials is $(x+a)$ and $12x^2(x+a)(x^2-a^2)$ respectively. Find the product of the two polynomials.

$$HCF = (x+a)$$
$$LCM = 12x^2(x+a)(x^2-a^2)$$

Let $p(x) \times q(x)$ be the product of two polynomials

By using formula

$$p(x) \times q(x) = HCF \times LCM$$
$$p(x) \times q(x) = [(x+a)][12x^2(x+a)(x^2-a^2)]$$
$$p(x) \times q(x) = [(x+a)][12x^2(x+a)(x+a)(x-a)]$$
$$\boldsymbol{p(x) \times q(x) = 12x^2(x+a)^3(x-a)}$$